# 使用统计信息通过 SMB 监控 Hyper-V 和 SQL Server 活动

ONTAP 9

NetApp
June 28, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/smb-hyper-v-sql/determine-statistics-objects-counters-task.html on June 28, 2022. Always check docs.netapp.com for the latest.

# 目录

使用统计信息通过 SMB 监控 Hyper-V 和 SQL Server 活动 . . . . . 1
确定可用的统计信息对象和计数器 . . . . . 1
显示 SMB 统计信息 . . . . . 4

# 使用统计信息通过 SMB 监控 Hyper-V 和 SQL Server 活动

## 确定可用的统计信息对象和计数器

在获取有关 CIFS ， SMB ，审核和 BranchCache 哈希统计信息以及监控性能的信息之前，您必须了解哪些对象和计数器可用于获取数据。

步骤

1. 将权限级别设置为高级：

   ` * 设置 -privilege advanced*`

2. 执行以下操作之一：

| 要确定的内容 | 输入 ... |
|---|---|
| 哪些对象可用 | ` * 统计信息目录对象 show*` |
| 可用的特定对象 | ` * 统计信息目录对象 show object *object_name**` |
| 哪些计数器可用 | ` * 统计信息目录计数器 show object *object_name**` |

   有关哪些对象和计数器可用的详细信息，请参见手册页。

3. 返回到管理权限级别：

   ` * 设置 -privilege admin*`

以下命令显示与集群中的 CIFS 和 SMB 访问相关的选定统计信息对象的说明，如高级权限级别所示：

```
cluster1::> set -privilege advanced

Warning: These advanced commands are potentially dangerous; use them only
when directed to do so by support personnel.
Do you want to continue? {y|n}: y

cluster1::*> statistics catalog object show -object audit
    audit_ng                    CM object for exporting audit_ng
performance counters

cluster1::*> statistics catalog object show -object cifs
    cifs                        The CIFS object reports activity of the
                                Common Internet File System protocol
                                ...

cluster1::*> statistics catalog object show -object nblade_cifs
    nblade_cifs                 The Common Internet File System (CIFS)
                                protocol is an implementation of the
Server
                                ...

cluster1::*> statistics catalog object show -object smb1
    smb1                        These counters report activity from the
SMB
                                revision of the protocol. For information
                                ...

cluster1::*> statistics catalog object show -object smb2
    smb2                        These counters report activity from the
                                SMB2/SMB3 revision of the protocol. For
                                ...

cluster1::*> statistics catalog object show -object hashd
    hashd                       The hashd object provides counters to
measure
                                the performance of the BranchCache hash
daemon.
cluster1::*> set -privilege admin
```

以下命令显示有关在高级权限级别看到的 `CIFS` 对象的某些计数器的信息：

> 此示例不会显示 `cifs` 对象的所有可用计数器；输出将被截断。

```
cluster1::> set -privilege advanced

Warning: These advanced commands are potentially dangerous; use them only
when directed to do so by support personnel.
Do you want to continue? {y|n}: y

cluster1::*> statistics catalog counter show -object cifs

Object: cifs
    Counter                     Description
    ---------------------------
----------------------------------------------
    active_searches             Number of active searches over SMB and
SMB2
    auth_reject_too_many        Authentication refused after too many
                                requests were made in rapid succession
    avg_directory_depth         Average number of directories crossed by
SMB
                                and SMB2 path-based commands
    ...                         ...

cluster2::> statistics start -object client -sample-id
Object: client
    Counter                                                     Value
    -------------------------------- --------------------------------
    cifs_ops                                                        0
    cifs_read_ops                                                   0
    cifs_read_recv_ops                                              0
    cifs_read_recv_size                                            0B
    cifs_read_size                                                 0B
    cifs_write_ops                                                  0
    cifs_write_recv_ops                                             0
    cifs_write_recv_size                                           0B
    cifs_write_size                                                0B
    instance_name                             vserver_1:10.72.205.179
    instance_uuid                                     2:10.72.205.179
    local_ops                                                       0
    mount_ops                                                       0


[...]
```

# 显示 SMB 统计信息

## 您可以显示各种 SMB 统计信息来监控性能和诊断问题。

步骤

1. 使用 `satictics start` 和可选的 `satictics stop` 命令收集数据样本。

   有关这些命令的详细信息，请参见 "《系统管理参考》"。

2. 执行以下操作之一：

| 要显示统计信息的对象 | 输入以下命令 ... |
|---|---|
| SMB 的所有版本 | ` * statistics show -object cifs*` |
| SMB 1.0 | ` * statistics show -object smb1*` |
| SMB 2.x 和 SMB 3.0 | ` * statistics show -object smb2*` |
| 节点的 CIFS 子系统 | ` * statistics show -object nblade_cifs*` |

有关详细信息，请参见手册页。

版权信息

版权所有©2022 NetApp、Inc.。保留所有权利。Printed in the U.S.版权所涵盖的本文档的任何部分不得以任何形式或任何手段复制、包括影印、录制、 磁带或存储在电子检索系统中—未经版权所有者事先书面许可。

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

本软件由NetApp按"原样"提供、不含任何明示或默示担保、包括但不限于适销性和特定用途适用性的默示担保、特此声明不承担任何责任。IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice.NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp.The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S.patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

商标信息

NetApp、NetApp标识和中列出的标记 http://www.netapp.com/TM 是NetApp、Inc.的商标。其他公司和产品名称可能是其各自所有者的商标。